Wearable-Based Digital Biomarkers for Parkinson's Disease

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1 Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized by motor symptoms such as tremor, rigidity, and bradykinesia Shawen et al. (2021). Currently, diagnosis and severity assessment rely primarily on periodic clinical examinations and subjective rating scales, such as the MDS-UPDRS or the Hoehn and Yahr (H&Y) scale Mittal et al. (2021). However, these evaluations can be infrequent and subject to inter-rater variability Shawen et al. (2021).

[0]Project repository on GitHub: `https://github.com/PaoloFabbri8/Behavioral-Health-Informatic-Project.git`

In this context, wearable technologies, such as smartwatches, provide a powerful alternative by enabling continuous physiological data collection in real-world settings Shawen et al. (2021). Recent literature highlights that the use of inertial sensors (accelerometers and gyroscopes) combined with machine learning algorithms can achieve accuracies exceeding 85% in symptom detection El Maachi et al. (2021). While accelerometers are considered the standard for characterizing human activity, the inclusion of gyroscopes can significantly improve bradykinesia detection Shawen et al. (2021).

Nevertheless, a critical trade-off exists between data complexity and the computational resources of wearable devices, such as battery life Shawen et al. (2021). Moreover, PD diagnosis is complex, requiring the assessment of both motor and non-motor symptoms El Maachi et al. (2021). Many studies have focused exclusively on movement signals, but integrating multimodal data,including clinical questionnaires and demographic information, may provide a more comprehensive view and increase predictive model reliability Mittal et al. (2021).

This project aims to address this gap by analyzing which digital biomarkers, derived from both sensor and clinical data from the PADS dataset (Parkinson's Disease Smartwatch Dataset), are most effective in identifying patients at risk of developing the disease.

1.1 Research Question

Starting from an initial exploratory phase of the dataset and a preliminary analysis of its composition, it was possible to identify three main research questions guiding this work.

Following this, the three research questions were developed:

- **RQ1.** Can machine learning distinguish Parkinson disease from other movement disorders using smartwatch sensor data?
- **RQ2.** Which smartwatch-derived digital biomarkers are most predictive of Parkinson disease?
- **RQ3.** Do smartwatch-based digital biomarkers improve early screening for Parkinson disease compared to self-reported symptoms?

The analysis was structured on two complementary levels. On the one hand, an in-depth analysis of the raw data was conducted in order to highlight relevant patterns, anomalies, and distributions. On the other hand, considerable attention was devoted to descriptive statistics and feature engineering techniques, with the aim of outlining a clearer profile of the research and what was intended to be effectively developed and built.

2 Exploratory Data Analysis (EDA)

The initial phase of the data analysis focused on loading and integrating the three main data sources available: questionnaire data, time-series data collected from smartwatch-based tests, and patients' demographic information.

We began by examining the demographic dataset, with particular attention to missing values. Overall, the data quality was high, with only one variable, `appearance_in_first_`

`grade_kinship`, containing missing values. This variable was subsequently excluded from the analysis, as it was not considered relevant for predictive purposes.

The dataset encompasses six distinct medical conditions, with Parkinson's Disease (PD) representing the majority class (58.8%), followed by healthy controls (16.8%) and other movement disorders (12.8%). Less frequent conditions include Essential Tremor (6.0%), Atypical Parkinsonism (3.2%), and Multiple Sclerosis (2.3%).

This highlights the presence of class imbalance, with PD being substantially overrepresented compared to rarer conditions, which together account for only a small fraction of the sample.

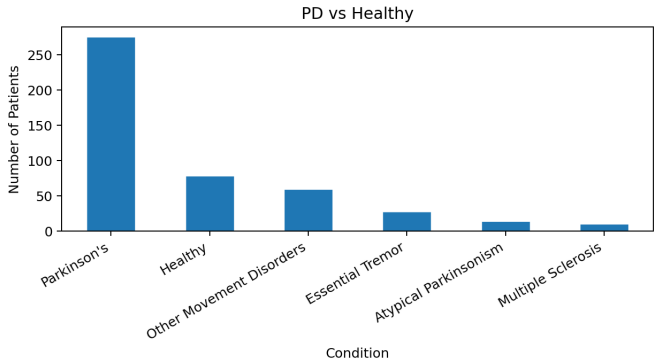

Figure 1: *Barplot of the number of patients with PD vs Healthy*

An ANOVA test was then conducted to assess differences in age across groups, revealing a non-uniform distribution ($p = 0.00117$), with the Parkinson's cohort showing one of the highest median ages. This pattern is likely attributable to the progressive nature of the disease; early symptoms are often subtle or misidentified, leading to a formal diagnosis only in later life as neurodegeneration advances.

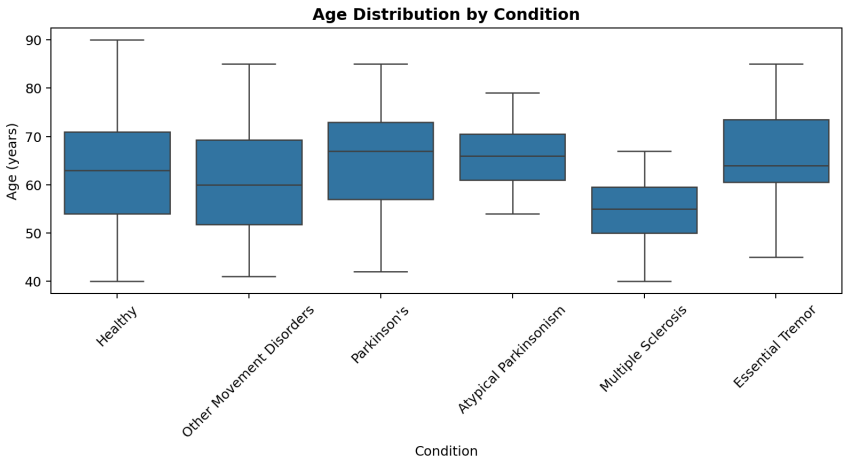

Figure 2: *Boxplot of the age distribution by condition*

Subsequently, we focused on the gender distribution. While women generally exhibit higher life expectancy, the marked male predominance in the Parkinson's cohort ($p < 0.001$) suggests that environmental and occupational factors may play a decisive role. Given that the sample consists of individuals over the age of 50, this gender gap could be linked to

historical differences in professional exposure. Men of this generation were more likely to be employed in strenuous manual labor or industries involving exposure to neurotoxic substances (such as pesticides or industrial chemicals), which are recognized risk factors for the development of Parkinson's disease and essential tremors.

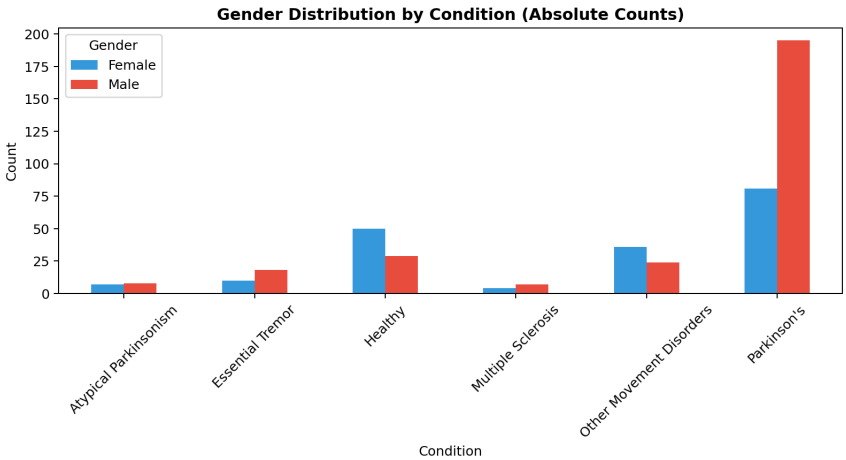

Figure 3: *Barplot of the gender distribution by condition*

The analysis of body weight across the different conditions yielded a statistically significant result through a one-way ANOVA test ($F = 2.439, p = 0.0337$). While the p-value falls below the standard threshold of 0.05, the significance is relatively marginal compared to other demographic variables. The higher body weight observed in Parkinson's patients could likely be attributed to the motor challenges associated with the condition, such as tremors and mobility difficulties. These symptoms often hinder regular physical activity, leading patients toward a more sedentary lifestyle, which in turn contributes to weight gain.

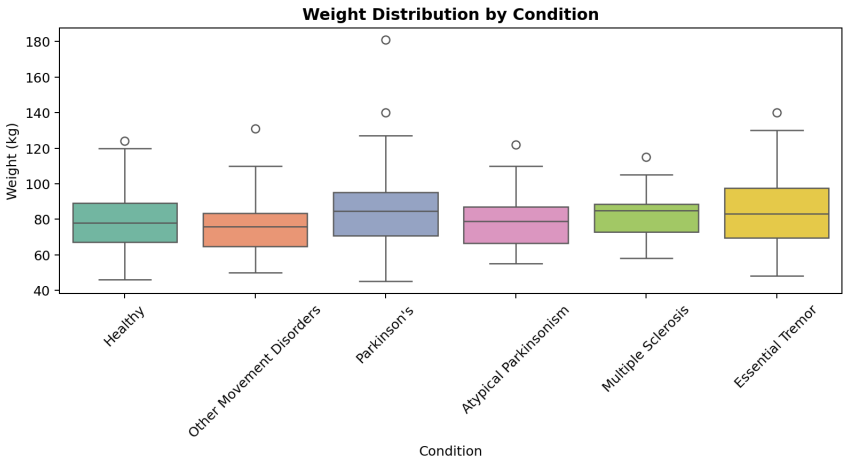

Figure 4: *Barplots of weight distibution by condition*

Following the initial demographic and distribution analysis, we focused on the time-series data collected via smartwatches. To identify which digital biomarkers exhibit the most significant discriminatory power between PD and Non-PD subjects, a Welch's t-test was performed. This univariate analysis, summarized in Table 1, serves to evaluate the

mean differences in motion and frequency features, providing a statistical baseline for the subsequent development of predictive models.

The results indicate that only two features remain statistically significant at the $p < 0.05$ level: tremor band power (4-6 Hz) ($p = 0.008$) and mean accelerometer magnitude ($p = 0.044$). Interestingly, the tremor band power is higher in the PD group (0.0017 vs 0.0013), confirming the presence of pathological rest tremor. Conversely, the acceleration mean is lower in PD patients, which may be an objective indicator of bradykinesia (slowness of movement). However, it should be noted that after applying the False Discovery Rate (FDR) correction, these features no longer reach the significance threshold ($p_fdr > 0.05$). This suggests that while these biomarkers show promising trends, the high overlap between PD and Non-PD populations makes individual feature-based classification challenging without more complex machine learning approaches.

Table 1: *Welch's t-test Results: Parkinson's (PD) vs. Non-Parkinson's (Non-PD)*

Feature	**Mean PD**	**Mean Non-PD**	**t-stat**	**p-value**	**p-FDR (BH)**	**Signif.**
tremor_band_4_6Hz	0.0017	0.0013	2.663	0.008	0.054	ns
acc_mag_mean	0.1232	0.1302	-2.011	0.044	0.155	ns
acc_mag_rms	0.1533	0.1606	-1.787	0.074	0.172	ns
gyro_mag_std	0.5533	0.5727	-1.452	0.147	0.256	ns
acc_mag_std	0.0864	0.0889	-1.124	0.261	0.366	ns
dominant_freq_hz	0.7384	0.7622	-0.653	0.514	0.599	ns
gyro_mag_mean	0.7234	0.7326	-0.460	0.645	0.645	ns

BH: Benjamini-Hochberg correction; ns: not significant after FDR correction ($p > 0.05$).

The same statistical approach was applied to the survey responses. In this case, the vast majority of items demonstrated significant differences between the two groups, maintaining their significance even after FDR correction. This high level of discrimination is likely due to the nature of the questions, which are specifically designed to capture early symptoms and pathological signs. Unlike the raw sensor data from smartwatches, which can be influenced by daily variability, the structured survey responses act as a direct clinical proxy for the disease state, thus providing a clearer separation between the Parkinson's and Non-Parkinson's cohorts.

Table 2: *Full Per-Item Significance Summary: Parkinson's vs. Non-Parkinson's*

Item ID	n_PD	n_Non-PD	p-value	p-FDR (BH)	Signif. (FDR)
01	276	193	1.96e-10	1.47e-09	
02	276	193	2.93e-17	8.80e-16	
03	276	193	2.55e-02	3.33e-02	
04	276	193	3.96e-01	4.24e-01	ns
05	276	193	7.61e-09	3.81e-08	
06	276	193	1.00e+00	1.00e+00	ns
07	276	193	1.33e-06	5.69e-06	
08	276	193	8.93e-14	1.34e-12	
09	276	193	4.15e-10	2.49e-09	
10	276	193	1.11e-03	2.08e-03	
11	276	193	2.77e-02	3.41e-02	
12	276	193	2.61e-05	8.71e-05	
13	276	193	2.84e-02	3.41e-02	
14	276	193	8.81e-05	2.40e-04	
15	276	193	2.30e-04	4.93e-04	
16	276	193	4.85e-02	5.60e-02	ns
17	276	193	5.79e-01	5.99e-01	ns
18	276	193	2.33e-02	3.33e-02	
19	276	193	1.78e-04	4.11e-04	
20	276	193	9.15e-03	1.53e-02	
21	276	193	6.93e-04	1.39e-03	
22	276	193	4.10e-05	1.23e-04	
23	276	193	2.29e-01	2.55e-01	ns
24	276	193	1.44e-05	5.41e-05	
25	276	193	5.30e-12	5.30e-11	
26	276	193	9.94e-05	2.48e-04	
27	276	193	2.48e-02	3.33e-02	
28	276	193	1.99e-03	3.51e-03	
29	276	193	1.34e-02	2.01e-02	
30	276	193	1.31e-02	2.01e-02	

BH: Benjamini-Hochberg correction; ns: not significant after FDR correction ($p > 0.05$).

3 Refining the Plan

To develop a robust predictive framework, we implemented a multi-stage feature engineering pipeline.

3.1 Data Load and Merging

The integration process involved merging three heterogeneous data sources (sensor data, questionnaires, and demographic information) using a unique subject identifier in order to construct a comprehensive patient-level dataset.

First, the time series derived from accelerometer and gyroscope sensors were subjected to a structured preprocessing and feature extraction pipeline. An integrated dataset was constructed by linking each motion file to patient information and the corresponding clinical condition (Parkinson's disease / non-Parkinson's) through merge operations with the patient table. The raw signals, acquired along the three axes, were converted into numerical format, handling any invalid values as missing data. Subsequently, the observations were temporally ordered, and missing values were imputed using linear interpolation, complemented by forward and backward filling techniques.

As suggested in the dataset "Usage Notes", the first 0.5 seconds of each recording were removed because the smartwatches produced a vibration notification at the beginning. Signal denoising was then performed by applying a low-pass Butterworth filter (cutoff at 20 Hz) to improve the robustness of time-domain analyses. From the filtered signals, statistical features were extracted for each axis and for the overall magnitude, including mean, standard deviation, root mean square, and range. In addition, frequency-domain features were derived from the accelerometer magnitude, such as dominant frequency, total power, and power within the tremor-related band (4–6 Hz), obtained through a dedicated band-pass filter.

Extracted features were first aggregated at file level and then summarized at patient level by grouping by `subject_id` and condition, computing mean and standard deviation across repeated tests. This preserves intra-patient variability while producing a compact tabular representation.

Questionnaire responses (30 binary items) were reshaped into a subject-by-feature matrix. Subject identifiers were normalized across all sources to ensure consistent joins.

Demographic variables were also merged at this stage; however they were used only to support age-aware train/test stratification and were removed before feature selection and model training.

Finally, the binary target variable was defined from clinical condition: observations were labeled positive if the condition contained parkinson, and negative otherwise.

3.2 Train Test Splitting and feature selection

A stratified train/test split was performed at the patient level (80% training, 20% test) to ensure a realistic evaluation and prevent data leakage. Splitting by `subject_id` guarantees that all observations from the same patient are assigned exclusively to either the training or test set.

To improve comparability between splits, we used an age-aware stratification strategy: stratification was performed on the joint distribution of class label and age bins (with fallback to class-only stratification when needed). In this way, both disease prevalence and age distribution remain balanced across train and test sets.Johnson et al. (2024)

Although demographic variables were available in the merged table, they were not used as predictors. Specifically, age was retained only to drive age-aware stratification, while demographic variables were removed before feature selection and model training.

After analyzing the data and observing the results of feature selection, we noticed that when merging the datasets, questionnaire items tended to dominate as the most predictive features. This is likely because the questionnaire responses already serve as an initial screening for the disease, confirming the presence of symptoms that are strongly associated with Parkinson's.

This observation led us to reconsider the research question and refine it to explicitly investigate the differences in predictive power across data modalities. Specifically, we aim to understand whether features derived from sensor-based simulations can achieve comparable predictive performance to self-reported questionnaire items, and how combining both types of information can improve overall disease detection.

- **Updated Research Question (RQ3):** To what extent can smartwatch-based sensor features complement or replicate the predictive power of self-reported questionnaire data for early detection of Parkinson's disease, and how can the combination of both improve predictive performance?

To investigate the relative contribution of different data modalities, three patient-level datasets were created:

- **Dataset A:** sensor-derived features only
- **Dataset B:** questionnaire features only

- **Dataset C:** combined sensor + questionnaire features

We then applied a leakage-safe feature selection procedure on the training set to identify the most informative predictors for the model in all three datasets. First, features with more than 20% missing values were removed, and remaining missing entries were imputed using median values estimated exclusively from the training set. Next, multicollinearity was addressed by removing highly correlated features (correlation > 0.95), retaining those with greater discriminative power.

Subsequently, a bimodal statistical ranking was performed by combining ANOVA F-test and Mutual Information. Features were ranked according to a composite score derived from both metrics, and the top 25 features were selected for the final model.

All thresholds, imputation values, and statistical measures were computed exclusively on the training set and then applied to the test set, ensuring a fully leakage-free pipeline and an unbiased evaluation of model performance.

4 Model Data Analysis

4.1 Model Training and Evaluation

After constructing the three feature setups, the next step was to train and evaluate a predictive model for Parkinson's disease classification. As the main classifier, we selected Random Forest, which is widely used in biomedical and digital health literature because of its robustness, flexibility, and strong empirical performance on structured tabular data.

After constructing the three feature setups, the next step was to train and evaluate a predictive model for Parkinson's disease classification. As the main classifier, we selected Random Forest (RF), which is widely used in biomedical and digital health literature because of its robustness, flexibility, and strong empirical performance on structured tabular data.Shawen et al. (2021); Johnson et al. (2024)

Random Forest was chosen as the primary model due to its ability to capture non-linear relationships and complex interactions, its robustness to noise and multicollinearity in high-dimensional data, and its reasonable interpretability.

To optimize model performance, hyperparameter tuning was performed separately for each feature setup: sensor-only, questionnaire-only, and combined multimodal data. A Randomized Search with 5-fold cross-validation was used on the training set only, ensuring that no information from the test set influenced model selection. The main settings explored were: number of trees, maximum tree depth, minimum number of samples required for node splitting, minimum number of samples in leaf nodes, the number of features considered at each split, and class weighting.

Once the best hyperparameters had been identified, a final Random Forest model was trained for each setup using the full training split. The tuned models were then evaluated on the test set.

Model evaluation was based on multiple complementary metrics. Accuracy was used as a general measure of correct classification. The F1-score was included to summarize the balance between precision and recall, and ROC-AUC was used as the main discrimination metric.

Finally, a feature importance analysis was performed for each of the three datasets in order to quantify the contribution of individual predictors to model performance. This step was included to support model interpretability and to identify which variables were most influential in the classification process across the sensor-only, questionnaire-only, and combined setups.

5 Interpretation of results

After training the Random Forest models on each dataset using their respective optimal hyperparameters, we evaluated their performance on the test set using the metrics described above (Accuracy, ROC-AUC, and F1-score).

The results, summarized in the bar plot in Figure 5, reveal that the unimodal models achieve comparable accuracies (Sensors: 0.70; Questionnaires: 0.73), indicating that both data types contain significant predictive information. However, a more detailed look at the metrics highlights a more pronounced gap in terms of ROC-AUC, where the questionnaire-based model (0.82) demonstrates superior discriminative power compared to the sensor-only model (0.70). This suggests that the questionnaire features allow for a better global separation between Parkinson's patients and healthy controls across all possible decision thresholds.

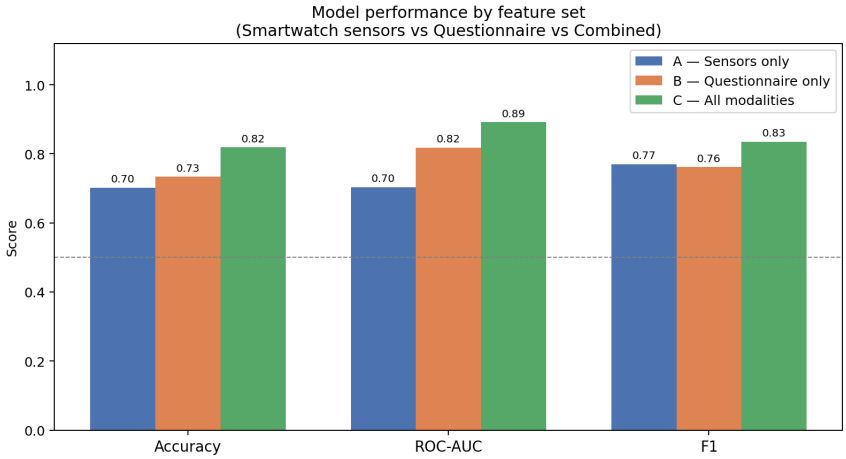

Figure 5: *Comparison of model performance across feature setups*

This high performance is largely due to the fact that questionnaire items are specifically designed to detect established clinical symptoms; if a patient is already able to report these symptoms, the predictive signal is naturally very strong.

In contrast, F1-score presents a slightly different trend, with the sensor model performing marginally better than the questionnaire one (0.77 vs 0.76). This discrepancy arises because, unlike ROC-AUC, which measures overall ranking quality, the F1-score evaluates the balance between precision and recall at a specific operational threshold.

Interestingly, as shown in Figure 6, only a small subset of these questions is necessary to achieve peak performance, with the F1-score for questionnaires reaching its maximum with just 5 features and declining as more variables are added. In contrast, the sensor-based model shows an upward trend as the number of features increases, suggesting that

objective behavioral data requires a broader set of signals to capture the complexity of motor impairment and improve the metric.

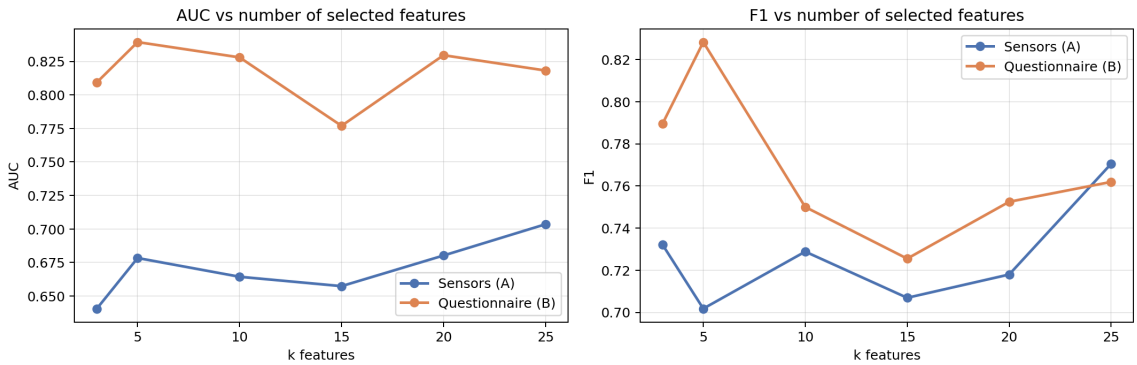

Figure 6: *Comparison of AUC and F1-score performance between Sensor and Questionnaire models*

Ultimately, when the two modalities are combined, a consistent and significant improvement is observed across all metrics. These findings strongly support the hypothesis of informational complementarity: while a few targeted questions can identify symptomatic patients, the addition of objective sensor data provides a richer, multi-dimensional representation of motor behavior, leading to a substantially more robust classification.

The potential of sensor-based data lies in its ability to provide a solid and objective foundation for early prevention. Unlike questionnaires, which, despite their effectiveness, are inherently susceptible to subjective bias, patient mood, and individual interpretation, sensors capture raw physiological signals that remain unaffected by the patient's personality or cognitive state.

Questionnaire responses can often be "noisy" or volatile, influenced by comorbid conditions or the patient's self-perception at a given moment. In contrast, wearable sensors offer consistent and verifiable metrics that track actual motor behavior. By leveraging these stable data points, we can move toward a model of continuous monitoring that is less prone to human error or misinterpretation. This objective accuracy is crucial for identifying subtle symptomatic changes that the patient might not yet perceive, thereby enabling more precise and timely preventive interventions in Parkinson's disease management.

5.1 Feature Importance

To understand which biological signals most influence the model, we analyzed the "Permutation Importance" on the test set. This technique identifies which features, if removed or altered, would cause the greatest drop in the model's predictive power.

As shown in Figure 7, the most dominant feature is the average sampling rate (`sampling_hz_mean`) which represents the average effective sampling rate of the smartwatch sensors during a recording session. While this might seem like a technical parameter, it could reflect how the micro-tremors characteristic of Parkinson's interact with the device's sensors. However, the most revealing finding comes from the remaining features in the top five.

All these features share a common trait: they measure variability across different sessions (identifiable by the `_std` suffix). Instead of measuring how fast or how much a patient

moves in a single moment, the model focuses on how much that movement changes and fluctuates from one recording to the next. Specifically:

- **Irregularity in Rotation and Acceleration:** Features like `gyro_mag_std_std`, `gyro_mag_std_mean`, and `acc_mag_std_std` track the inconsistency and average level of irregularity in rotational and linear movements. In healthy subjects, these patterns remain relatively stable; in Parkinson's patients, the intensity and "fluidity" of movement vary significantly across sessions.
- **Rotational Consistency:** The feature `gyro_mag_mean_std` further captures the cross-session inconsistency of rotational motion intensity, reflecting the difficulty in maintaining steady motor control.

In conclusion, 80% of the top five features are descriptors of cross-test variability. This confirms our core hypothesis: Parkinson's disease is best characterized not by a single "abnormal" signal, but by the loss of motor consistency. While a healthy individual produces stable and repeatable movement patterns, a Parkinson's patient exhibits high variability due to motor fluctuations and the progressive disruption of motor control circuitry.

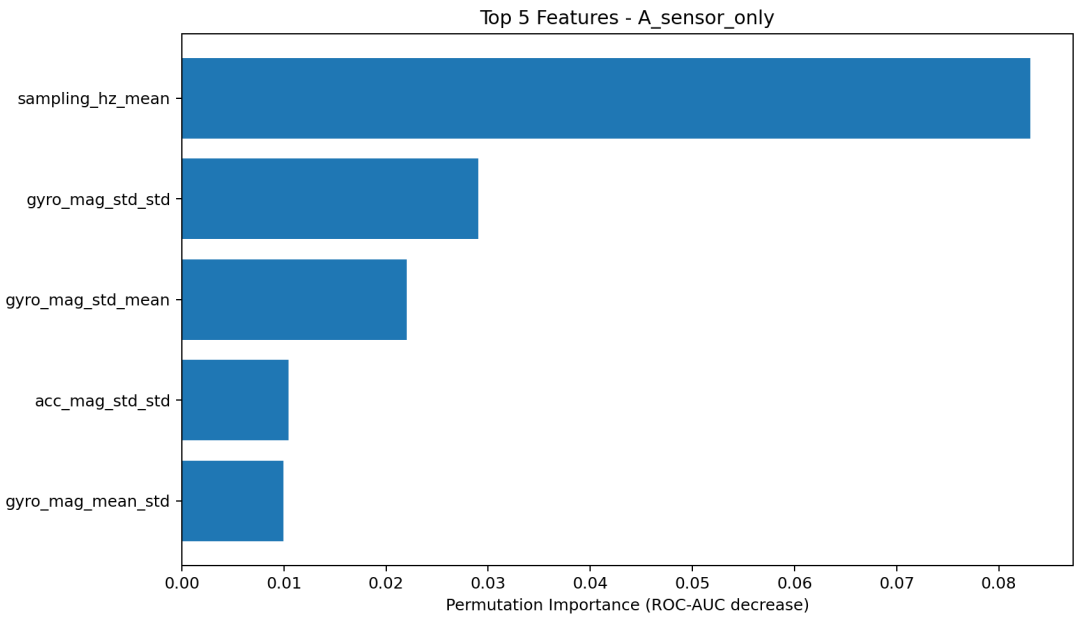

Figure 7: *Feature importance ranking for the sensor-only model. The prevalence of `_std` suffixes highlights how session-to-session variability is a primary predictor for Parkinson's disease.*

5.2 Conclusions

Returning to our research question, this study shows that smartwatch-derived sensor features can provide predictive performance that is broadly comparable to questionnaire-based features for Parkinson disease detection. The unimodal models were close in overall classification performance, with similar accuracy, while the questionnaire setup achieved higher ROC-AUC and the sensor setup achieved a slightly better F1-score. This pattern suggests that questionnaire features may offer stronger global ranking of risk, whereas

sensor features can be highly effective at the chosen operating threshold, especially for balancing precision and recall.

The strongest result is that combining both modalities produced the best performance across metrics. This supports the idea that sensor and questionnaire data are not redundant, but complementary. Questionnaire items capture explicit, symptom-level clinical information, while wearable sensors capture continuous motor behavior that may be difficult for patients to report directly. Integrating these two perspectives improves robustness and yields a more complete representation of disease-related patterns.

The feature-importance analysis further strengthens this interpretation. In the sensor-only setting, the most informative variables were largely related to variability across repeated tests, rather than single static signal levels. This indicates that fluctuations in movement dynamics are central for separating PD from non-PD participants. In practical terms, well-designed repeated motor tasks, combined with appropriate feature engineering, allow wearable data to provide measurements that are qualitatively comparable to questionnaire information, and in some decision contexts potentially as useful.

Overall, the project demonstrates that smart wearables are not only supportive tools, but promising predictive instruments for early screening and prevention-oriented monitoring of Parkinson disease. A realistic clinical direction is a hybrid pipeline in which passive or semi-structured sensor monitoring is paired with short targeted questionnaires.

6 Contributions

This project was entirely carried out independently. Throughout the process, I made use of AI-based tools such as ChatGPT and Gemini primarily to improve the clarity and quality of the written report, as well as to support the search for relevant materials and useful references. Additionally, I used Copilot as a coding assistant to help streamline the development process, particularly for writing code and organizing the project structure. All final decisions, implementations, and interpretations remain my own.

7 References

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